Step 5 : Calculation of Test statistic

ANOVA Table (one-way)

Source of variation	Sum of squares	Degrees of freedom	Mean sum of squares	F-ratio
Treatments	SST	k-1	$MST = \frac{SST}{k-1}$	$F_0 = \frac{MST}{MSE}$
Error	SSE	n-k	$MSE = \frac{SSE}{n-k}$	
Total	TSS	n-1		

Step 6 : Critical value

$f_e = f_{(k\text{-}1,\, n\text{-}k),\alpha}$.

Step 7 : Decision

If $F_0 < f_{(k\text{-}1,\, n\text{-}k),\alpha}$ then reject H_0.

3.3.3 Merits and Demerits of One-Way ANOVA

Merits

- Layout is very simple and easy to understand.
- Gives maximum degrees of freedom for error.

Demerits

- Population variances of experimental units for different treatments need to be equal.
- Verification of normality assumption may be difficult.

Example 3.4

Three different techniques namely medication, exercises and special diet are randomly assigned to (individuals diagnosed with high blood pressure) lower the blood pressure. After four weeks the reduction in each person's blood pressure is recorded. Test at 5% level, whether there is significant difference in mean reduction of blood pressure among the three techniques.

Medication	10	12	9	15	13
Exercise	6	8	3	0	2
Diet	5	9	12	8	4

Solution:

Step 1 : Hypotheses

Null Hypothesis: H_0: $\mu_1 = \mu_2 = \mu_3$

That is, there is no significant difference among the three groups on the average reduction in blood pressure.

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